7-2-2026

Prandtl boundary large equation

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{u}}{\partial y} = 0$$

$$\bar{u}\frac{\partial \bar{u}}{\partial x} + \bar{v}\frac{\partial \bar{u}}{\partial y} = \nu \frac{\partial^2 \bar{u}}{\partial y^2} \qquad \beta = \frac{1}{T_\infty}$$

$$\bar{u}\frac{\partial T}{\partial x} + \bar{v}\frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} + \underbrace{g\beta(T - T_\infty)}_{0} \qquad \boxed{\alpha = \frac{k}{\rho c_p}}$$

Turbulent flow

more chaotic

Richard Feynman

differences
- RANS – Reynolds Averaged Navier-Stokes
- LES – Large-Eddy simulation [turbulent]
- DNS – Direct Numerical Simulation [self sufficient]

$$\vec{u} = u\hat{i} + v\hat{j} + w\hat{k}$$

$$\text{curl}\,\vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} \quad ; \quad \omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Laminalization region [FFT]

$$\frac{\partial \bar{u}}{\partial \bar{x}} = \frac{\partial}{\partial x}\left(\frac{\nu}{L} Gr^{-1/2} u\right) \cdot \frac{\partial x}{\partial \bar{x}}$$

$$= \frac{\nu}{L} Gr^{-1/2} \frac{\partial u}{\partial x} \cdot \frac{1}{L} \quad \left[\frac{\partial}{\partial x}\left(\frac{\bar{x}}{L}\right) = \frac{1}{L}\right]$$

$$\frac{\partial \bar{u}}{\partial \bar{x}} = \frac{\nu}{L^2} Gr^{-1/2} \frac{\partial u}{\partial x}$$

$$\frac{\partial \bar{v}}{\partial \bar{y}} = \frac{\partial}{\partial y}\left(\frac{\nu}{L} Gr^{-1/4} \cdot v\right) \cdot \frac{\partial y}{\partial \bar{y}}$$

$$= \frac{\nu}{L} Gr^{-1/4} \cdot \frac{\partial v}{\partial y} \cdot Gr^{1/4} \cdot \frac{1}{L}$$

$$= \frac{\nu}{L} Gr^{1/2} \frac{\partial v}{\partial y}$$

$$\frac{\partial \bar{u}}{\partial \bar{y}} = \frac{\nu}{L^2} Gr^{3/4} \frac{\partial u}{\partial y}$$

$$\frac{\partial^2 \bar{u}}{\partial \bar{y}^2} = \frac{\nu}{L^3} Gr \frac{\partial^2 u}{\partial y^2}$$

Now, $\bar{u} \frac{\partial \bar{u}}{\partial \bar{x}} + v \frac{\partial \bar{u}}{\partial \bar{y}} = \nu \frac{\partial^2 \bar{u}}{\partial \bar{y}^2}$

$$\Rightarrow \frac{\nu}{L} Gr^{1/2} u \cdot \frac{\nu}{L^2} Gr^{1/2} \frac{\partial u}{\partial x} + \frac{\nu}{L}(\wedge) \frac{\nu}{L^2} Gr^{3/4} \cdot \frac{\partial u}{\partial y} = \nu \cdot \frac{\nu}{L^3} Gr \frac{\partial^2 u}{\partial y^2}$$

$$\Rightarrow \frac{\nu^2}{L^3} Gr \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}\right) = \frac{\nu^2}{L^3} Gr \frac{\partial^2 u}{\partial y^2}$$

$$\Rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial y^2}$$

$$\theta = \frac{T - T_A}{\Delta T}$$

$$\Rightarrow T = T_\infty + \Delta T \theta$$

$$\frac{\partial T}{\partial \bar{x}} = \frac{\partial}{\partial x}(T_\infty + \Delta T \theta)\frac{\partial x}{\partial \bar{x}}$$

$$= \frac{\Delta T}{L} \cdot \frac{\partial \theta}{\partial x}$$

$$\frac{\partial T}{\partial \bar{y}} = \frac{\partial}{\partial y}(T_\infty + \Delta T \theta)\frac{\partial y}{\partial \bar{y}}$$

$$= \frac{\Delta T}{L} \cdot Gr^{1/4} \cdot \frac{\partial \theta}{\partial y}$$

$$\frac{\partial^2 T}{\partial \bar{y}^2} = \frac{Gr^{1/2}}{L^2} \Delta T \frac{\partial^2 \theta}{\partial y^2}$$

$$g\beta(T - T_\infty) = g\beta \Delta T \theta$$

$$= g\beta \frac{Gr L^3}{\nu^2} \cdot \frac{1}{Gr} \theta$$

$$= \theta$$

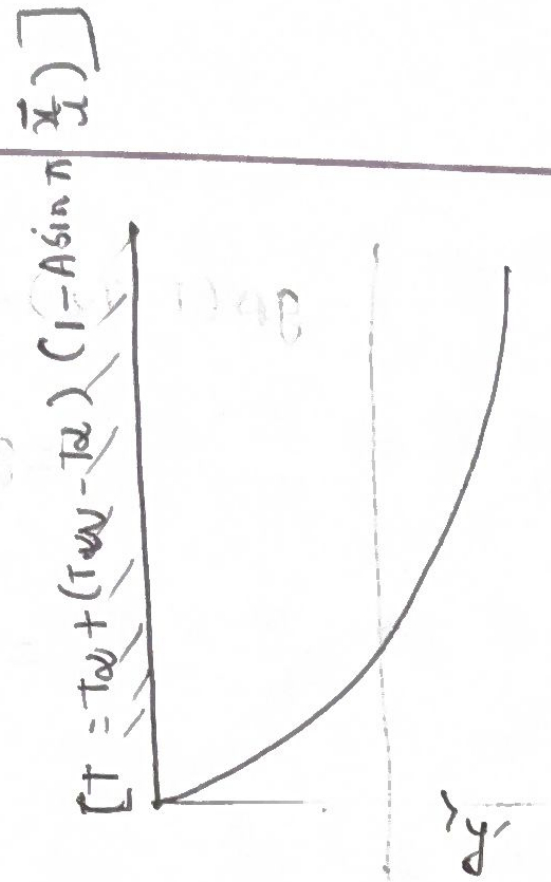
Now, $\bar{u}\frac{\partial T}{\partial \bar{x}} + \bar{v}\frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$

$$\Rightarrow \frac{\nu}{L} Gr^{1/2} u \cdot \cancel{\frac{\nu}{L^2} Gr^{1/2}} \frac{\Delta T}{L} \cdot \frac{\partial \theta}{\partial x} + \frac{\nu}{L} \cdot \cancel{\frac{\nu}{L^2}} Gr^{1/4} \cdot v \frac{Gr^{1/4}}{L} \cdot \Delta T \cdot \frac{\partial \theta}{\partial y}$$

$$= \alpha \cdot \frac{Gr^{1/2}}{L^2} \cdot \Delta T \frac{\partial^2 \theta}{\partial y^2}$$

$$\Rightarrow u \cdot \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} = \alpha \cdot \frac{1}{\nu} \frac{\partial^2 \theta}{\partial y^2}$$

$$= \frac{1}{Pr} \frac{\partial^2 \theta}{\partial y^2}$$



$T \to \infty$ as $y \to \infty$

$u \to 0$, $v \to \infty$

$T = T_\infty + (T_w - T_\infty)(1 + A \sin(\pi \frac{x}{l})$

$\bar{u} = \bar{v} = 0 =$ no slip condition

$T - T_\infty = (T_w - T_\infty)(1 + A \sin(\pi x)]$

$\to \dfrac{T - T_\infty}{T_w - T_\infty} = 1 + A \sin \pi x$

$\to \theta = 1 + A \sin(\pi x)$ at $y = 0$

$A\,(max) = 0.3$

$\left.\begin{array}{l}\theta = 0 \\ u \to 0\end{array}\right\}$ as $y \to \infty$

$\psi =$ stream function

$\psi = x^{3/4} f(x, \eta)$, $\eta = x^{-1/4} y$, $\theta = (x, \eta)$

$u = \dfrac{\partial \psi}{\partial y}$, $v = \dfrac{\partial \psi}{\partial x}$

$= x^{3/4} \dfrac{\partial f}{\partial \eta} \dfrac{\partial \eta}{\partial y}$

$= x^{3/4} \cdot \dfrac{\partial f}{\partial \eta} \cdot x^{-1/4}$

$= x^{1/2} \cdot \dfrac{\partial f}{\partial \eta}$

$\dfrac{\partial u}{\partial y} = x^{1/2} \cdot x^{-1/4} \cdot \dfrac{\partial^2 f}{\partial \eta^2}$

$\dfrac{\partial^2 u}{\partial y^2} = \dfrac{\partial^3 f}{\partial \eta^3}$

$$\frac{\partial \eta}{\partial x} = -\frac{1}{4}x^{-5/4}.y = -\frac{1}{4x}.x^{-1/4}y = -\frac{\eta}{4x}$$

$$v = -\frac{\partial \psi}{\partial x} = -\left\{\left[x^{3/4}\left(\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial x} + \frac{\partial f}{\partial \eta}\cdot\frac{\partial \eta}{\partial x}\right)\right] + \left[f\cdot\frac{3}{4}\cdot x^{-1/4}\right]\right\}$$

$$= -x^{3/4}\cdot\frac{\partial f}{\partial x} - \frac{1}{4x}\cdot\eta\frac{\partial f}{\partial \eta} + \frac{3}{4}x^{-1/4}\cdot f$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial y^2}$$

Now, $\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left(x^{\frac{1}{2}}\cdot\frac{\partial f}{\partial \eta}\right)$

$$\cancel{x^{\frac{1}{2}}\frac{\partial f}{\partial \eta}} = x^{\frac{1}{2}}\left(\frac{\partial^2 f}{\partial x\,\partial \eta} + \frac{\partial^2 f}{\partial \eta^2}\cdot\frac{\partial \eta}{\partial x}\right) + \frac{1}{2}x^{-\frac{1}{2}}\frac{\partial f}{\partial \eta}$$

$$= x^{\frac{1}{2}}\left(\frac{\partial^2 f}{\partial x\,\partial \eta} - \frac{1}{4x}\eta\frac{\partial^2 f}{\partial \eta^2}\right) + \frac{1}{2}x^{-\frac{1}{2}}\frac{\partial f}{\partial \eta}$$

$$\frac{\partial \theta}{\partial x} = \frac{\partial \theta}{\partial x}\cdot\frac{\partial x}{\partial x} + \frac{\partial \theta}{\partial \eta}\cdot\frac{\partial \eta}{\partial x}$$

$$= \frac{\partial \theta}{\partial x} - \frac{1}{4x}\cdot\eta\frac{\partial \theta}{\partial \eta}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial y^2}$$

$$\Rightarrow x^{\frac{1}{2}}\frac{\partial f}{\partial \eta}\left[x^{\frac{1}{2}}\frac{\partial^2 f}{\partial x\,\partial \eta} - \cancel{\frac{x^{\frac{1}{2}}}{4x}\eta\frac{\partial^2 f}{\partial \eta^2}}^{\,0} + \frac{1}{2}x^{-\frac{1}{2}}\frac{\partial f}{\partial \eta}\right]$$

$$\left(-x^{3/4}\frac{\partial f}{\partial x} + \cancel{\frac{x^{3/4}}{4x}\eta\frac{\partial f}{\partial \eta}}^{\,0} + \frac{3}{4}x^{-\frac{1}{4}}f\right)x^{1/4}\frac{\partial^2 f}{\partial \eta^2} = \frac{\partial^3 f}{\partial x^3} + \theta$$

$$\frac{\partial f}{\partial \eta} = f' \quad ; \quad \frac{\partial^2 f}{\partial \eta^2} = f'' \quad , \quad \frac{\partial^3 f}{\partial \eta^3} = f'''$$

$$\Rightarrow x \frac{\partial f}{\partial \eta} \cdot \frac{\partial^2 f}{\partial \eta \partial x} + \frac{1}{2}\left(\frac{\partial f}{\partial \eta}\right)^2 - x \frac{\partial f}{\partial x} \cdot \frac{\partial f}{\partial \eta^2} - \frac{3}{4} f \frac{\partial^2 f}{\partial \eta^2} = \frac{\partial^3 f}{\partial \eta^3} + \theta$$

$$\Rightarrow \frac{\partial^3 f}{\partial \eta^3} + \frac{3}{4} f \frac{\partial^2 f}{\partial \eta^2} - \frac{1}{2}\left(\frac{\partial f}{\partial \eta}\right)^2 + \theta \cancel{\left[\left(\frac{\partial f}{\partial x} \cdot \frac{\partial^2 f}{\partial x^2}\right) - \frac{\partial f}{}\right]}$$

$$= x\left(\frac{\partial f}{\partial \eta} \cdot \frac{\partial^2 f}{\partial x \cdot \partial \eta} - \frac{\partial f}{\partial x} \frac{\partial^2 f}{\partial \eta^2}\right)$$

$$\Rightarrow f''' + \frac{3}{4} f \cdot f'' - \frac{1}{2}(f')^2 + \theta = x\left(f' \cdot \frac{\partial f'}{\partial x} - f'' \frac{\partial f}{\partial x}\right)$$

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$$x = \xi$$

$$\theta = \theta(x, \eta) = \theta(\xi, \eta)$$

$$\frac{\partial \theta}{\partial x} = \frac{\partial \theta}{\partial \xi} \cdot \frac{\partial \xi}{\partial \eta} + \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x}$$

$$= \frac{\partial \theta}{\partial \xi} \cdot 1 \pm \frac{1}{4x} \eta \theta'$$

$$= \frac{\partial \theta}{\partial x} - \frac{1}{4x} \eta \theta'$$

$$\frac{\partial \theta}{\partial y} = \frac{\partial \theta}{\partial x} \cdot \frac{\partial x}{\partial y} + \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y}$$

$$= x^{-\frac{1}{4}} \cdot \theta'$$

$$\frac{\partial^2 \theta}{\partial y^2} = x^{-\frac{1}{2}} \theta''$$

$$u\frac{\partial \theta}{\partial x} + u\frac{\partial \theta}{\partial y} = \frac{1}{\rho r}\cdot\frac{\partial^2 \theta}{\partial y^2} \qquad x^{-\frac{1}{4}}\theta'$$

$$\Rightarrow x^{\frac{1}{2}} f'\left(\frac{\partial \theta}{\partial x} - \frac{1}{4x}\eta\theta'\right) + \left(-x^{\frac{3}{4}}\cdot\frac{\partial f}{\partial \eta} + \frac{x^{3/4}}{4x}\eta f' - \frac{3}{4}x^{-\frac{1}{4}} f\right)\theta'$$

$$= x^{-\frac{1}{2}}\theta''$$

$$\Rightarrow x f'\frac{\partial \theta}{\partial x} - x\frac{\partial f}{\partial x}\theta' - \frac{3}{4} f\theta' = \theta''$$

$$\therefore \theta'' + \frac{3}{4} f\theta' = x\left(f'\frac{\partial \theta}{\partial x} - \theta'\frac{\partial f}{\partial x}\right)$$

B.C. $\quad f = 0 \;,\; \theta = 1+\alpha$

$f \to 0 \;,\; \theta \to 0 \quad$ as $\quad \eta \to \infty$

$$f' + \frac{3}{4} f f'' - \frac{1}{2} f'^2(\theta) = x\left(f'\frac{\partial f'}{\partial x} - f''\frac{\partial f}{\partial x}\right)$$